Monday Warm-Up: Unit 5, Electromagnetism

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1 Memory Bank

- $U_{\mathrm{C}} = \frac{1}{2}Q^2C^{-1}$ ... Energy stored in a capacitor.
- $C = \epsilon_0 A/d$ ... Parallel plate capacitance.
- $U_{\mathrm{C}} = \frac{1}{2}LI^2$ ... Energy stored in an inductor.
- $L = \mu_0 N^2 A/l$ ... Solenoidal inductance.
- $c = f\lambda$ ... The relationship between speed of light, c, the frequency, f, and the wavelength, λ.
- $c = 299,792,458$ m s^{-1}, or $c \approx 0.3$ m ns^{-1}.
- $v = c/n$ ... The actual speed of light, v in a material with index of refraction n.
- $\bar{S} = E^2/(2c\mu_0)$ ... Average **intensity**, or W m^{-2}, of electromagnetic energy.
- $\cos(x)\cos(y) = (1/2)(\cos(x+y)+\cos(x-y))$ ... Trigonometric identity for *mixing* sinusoids.

2 Electromagnetic Waves and Energy

1. (a) What is the intensity of a wave created by an E-field with magnitude 1000 V m^{-1} at the source? (b) How do you think the previous result scales with distance from the source?

2. (a) What is the wavelength of a 6 GHz microwave in vacuum? (b) What is the frequency of a wave with wavelength 3 cm in vacuum? (d) If a wave has a wavelength of 10 cm in air, what is the wavelength in ice if the index of refraction is $n = 1.78$?

3 Signal Mixing and AM Radio

1. (a) Suppose $f(t) = 4\cos(2\pi f_1 t)$ and $g(t) = 2\cos(2\pi f_2 t)$. Using a formula from the Memory Bank, calculate the frequencies present in the product $f(t)g(t)$. (b) Suppose $f_1 = 1440$ kHz, and $f_2 = 560$ Hz. What frequencies exist in $f(t)g(t)$? (c) Suppose an RLC resonator is built to take an audio signal at f_2 and mix it with a *carrier frequency* at f_1. The product is the *radiated* across space. Graph this radiated signal $f(t)g(t)$.

4 Geometric Optics

1. Suppose a laser beam reflects from a mirror. (a) If the incident angle with respect to the normal direction of the surface is 30 degrees, what is the reflected angle? (b) Suppose two mirrors meet in a corner at a 90 degree angle. At what incident angle does the incoming ray leave in a direction parallel to the incoming direction?

5 Wave Optics

1. Suppose a beam of optical light enters glass, and the incident angle with respect to the normal direction at the interface is not equal to the transmitted angle. If light were a stream of particles, rather than a wave, would that be possible?